Python Activity-Based Learning
17. Student Performance Trend Analysis

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

A student has marks from three examinations in one subject. Write a program to find the average mark and check whether the marks are improving, decreasing, or staying the same.

2. Objectives

• Read marks for three examinations.

• Calculate the average.

• Compare marks to identify the trend.

3. Assumptions

• Marks are from 0 to 100.

• Exactly three examination marks are entered.

• A higher later mark means improvement.

4. Algorithm (Step by Step)

1. Start.

2. Read marks from three examinations.

3. Check that each mark is between 0 and 100.

4. Calculate the average.

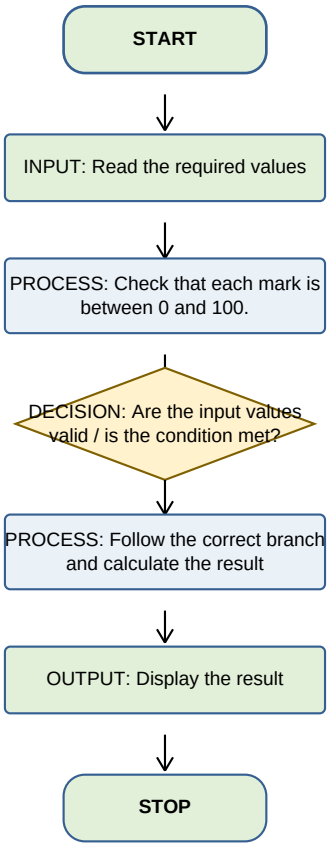
5. Compare the first, second, and third marks.

6. Display Improving, Decreasing, or No clear trend.

7. Display the average and stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT m1, m2, m3
IF any mark is outside 0 to 100 THEN PRINT Invalid marks
ELSE
    average = (m1 + m2 + m3) / 3
    IF m1 < m2 AND m2 < m3 THEN trend = Improving
    ELSE IF m1 > m2 AND m2 > m3 THEN trend = Decreasing
    ELSE trend = No clear trend
    PRINT average, trend
STOP
```

7. Python Source Code

```
m1 = float(input("Enter first exam mark: "))
m2 = float(input("Enter second exam mark: "))
m3 = float(input("Enter third exam mark: "))

if not (0 <= m1 <= 100 and 0 <= m2 <= 100 and 0 <= m3 <= 100):
    print("Invalid marks")
else:
    average = (m1 + m2 + m3) / 3

    if m1 < m2 < m3:
        trend = "Improving"
    elif m1 > m2 > m3:
        trend = "Decreasing"
    else:
        trend = "No clear trend"

    print("Average mark:", round(average, 2))
    print("Performance trend:", trend)
```

8. Test Cases

Input / Test	Expected result	What it checks
50, 60, 70	60.0; Improving	Marks rise each time

Input / Test	Expected result	What it checks
80, 70, 60	70.0; Decreasing	Marks fall each time
60, 65, 62	62.33; No clear trend	Marks do not move in one direction

9. Sample Output

```
Enter first exam mark: 50
Enter second exam mark: 60
Enter third exam mark: 70
Average mark: 60.0
Performance trend: Improving
```

10. Conclusion

This program calculates the average mark and uses conditions to describe the student's performance trend.